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Breaking the ice: A social exchange study on gamification, library anxiety, and the willingness to return at the University of Malta Library

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ABSTRACT

Purpose: Grounded in Social Exchange Theory, this study investigates the efficacy of gamified orientation as an initiatory social transaction to mitigate Library Anxiety. It aims to determine if gamification lowers the psychological "cost" of entry and increases the "benefit" of return intention compared to traditional methods, while evaluating the moderating role of "Game Literacy."

Design/methodology/approach: A quasi-experimental, post-test-only design was employed with 132 first-year undergraduates at the University of Malta. Following a pilot study to validate content, intact classes were assigned to either a traditional lecture ($n = 65$) or a gamified narrative quest ($n = 67$). Immediate affective outcomes were measured using the Library Anxiety Scale and Net Promoter Score.

Findings: Gamification produced a statistically significant reduction in Library Anxiety ($d = 1.55$) and higher return intentions compared to the control group. However, a significant interaction effect revealed a "literacy tax": while gamification benefited all participants, students with low game literacy reported higher anxiety than their high-literacy peers, indicating that game mechanics can impose a secondary cognitive load.

Originality/value: Unlike studies focusing solely on learning outcomes, this research quantifies the affective impact of gamification through a social exchange lens. It provides empirical evidence that while gamification successfully "breaks the ice," equitable implementation requires "low-threshold" mechanics to ensure non-gamers are not disadvantaged.

Introduction

For many first-year undergraduates, the academic library is less a sanctuary of learning and more an imposing institution of complex rules and unfamiliar systems. This phenomenon, widely documented as "Library Anxiety," creates an immediate barrier to access, often causing students to avoid the library entirely rather than risk the embarrassment of not knowing how to use it (McAfee, 2018; Mellon, 1986). Traditionally, libraries have attempted to mitigate this anxiety through orientation tours and lectures. However, these passive formats often fail to engage the "digital native" student, potentially reinforcing the perception of the library as a static, high-effort environment.

To understand the mechanics of this disengagement, this study adopts the framework of Social Exchange Theory (SET). As posited by Scicluna (2025), student engagement with library services is not automatic; it is a rational calculation where individuals weigh the "costs" of interaction against the "benefits." In a traditional orientation, the "cost" (boredom, cognitive load, social intimidation) often outweighs the perceived "benefit" (future information literacy). If the exchange feels unfair, the student simply won't come back.

Gamification, the application of game design elements in non-game contexts, offers a radical restructuring of this equation (Nah et al., 2014). By transforming the library induction into a quest, scavenger hunt, or narrative challenge, libraries aim to artificially lower the psychological cost of entry while boosting the immediate benefit through enjoyment and immersion (Triantafyllou et al., 2025). Theoretically, if the "cost" of anxiety is replaced by the "benefit" of play, the likelihood of voluntary return should increase.

However, the assumption that "games are fun for everyone" requires critical scrutiny. Recent findings by Durodolu et al. (2025) highlight the variable of "Game Literacy", a user's familiarity with game mechanics and logic. If a student lacks this literacy, a gamified orientation may inadvertently *increase* the cognitive cost, replacing library anxiety with "game anxiety." Therefore, this study argues that to justify gamification, we must measure its immediate impact on the user's affective state, not just their test scores.

This research employs a quasi-experimental, post-test-only design to compare the immediate psychological effects of gamified versus

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traditional library orientation. It seeks to determine if gamification successfully “breaks the ice” by lowering anxiety and increasing the willingness to return, while controlling for the moderating influence of the students' own gaming habits.

Research objectives

To address the effectiveness of gamification as an outreach strategy, this study pursues the following three objectives:

1. To Determine Affective Efficacy: To compare the levels of Library Anxiety (Perceived Cost) reported by students immediately following a gamified orientation versus those attending a traditional lecture-based tour.
2. To Measure the “Exchange” Value: To determine if gamified orientation leads to a higher Net Promoter Score (NPS) and intention to return (Perceived Benefit) compared to traditional methods.
3. To Evaluate the “Gamer” Bias: To investigate if Game Literacy moderates the relationship between the instructional method and anxiety reduction, determining if non-gamers are disadvantaged by the gamified approach.

Literature review

Gamification of library services: strategies, applications, and theoretical foundations

Gamification has proven to be a highly effective strategy for enhancing motivation and user engagement by integrating game elements into non-game contexts (Adeyemi et al., 2025; Felker & Phetteplace, 2014; Nah et al., 2014). The mechanics driving these outcomes are broadly illuminated by SET, whose principles provide a powerful framework for understanding gamified design not only in academic settings but across diverse external sectors like marketing, human resources (HR), and public welfare.

In marketing, fundamental research shows that gamified reciprocity and status-based rewards effectively utilize social exchange dynamics to boost cooperative consumer behavior and increase influence within user communities (Hoffman et al., 1998; Kamijo et al., 2020; Willer, 2009). These exchange-driven mechanics are similarly relevant to workforce management and the gig economy. For example, Cheng et al. (2020) used a spatial public goods game to demonstrate how targeted subsidies and productivity differentials can shape career choices, suggesting that gamified HR incentives can successfully guide workforce composition toward socially beneficial, public-oriented roles. Furthermore, in workforce management, SET has been integrated to assess how gamified platforms foster swift trust and reduce dropout rates among digital gig-economy workers by balancing the psychological costs of effort against the benefits of autonomy and achievement (Behl et al., 2021).

This alignment between individual incentives and public benefit naturally extends to broader public welfare initiatives. Research on club-based governance, such as water management clubs, highlights how grouping and trading designs decrease free-riding and improve the overall provision of public goods (Sağlam & Talbot-Jones, 2023). When viewed as gamified, club-like incentives, these governance designs are further strengthened by the monitoring and reward structures inherent in social exchange dynamics (Kamijo et al., 2020; Sağlam & Talbot-Jones, 2023).

Ultimately, this evidence from various fields illustrates that SET is fundamental to effective gamification. It leverages reciprocity, status rewards, sanctions, and group-based incentives to promote cooperative and mutually beneficial outcomes in markets, workplaces, and public goods provision.

In library science specifically, this approach transforms routine interactions and Information Literacy (IL) instruction into rewarding experiences using mechanics like points, badges, leaderboards, and narrative quests (Adedokun et al., 2021; Leenaraj et al., 2023). Current evidence shows that whether libraries use low-tech, design-thinking sessions or scalable Augmented Reality (AR) tours, gamified services consistently lead to positive results in learner engagement, digital literacy, and instructional confidence (Capdarest-Arest et al., 2019; T R & Gala, 2023; Yi-Shiang et al., 2013).

To understand the underlying efficacy of these gamified interventions, the literature increasingly advocates for a blended theoretical approach that synthesizes intrinsic motivation and exchange-based perspectives. Grounded in Self-Determination Theory (SDT) and cognitive evaluation, gamification is understood to scaffold intrinsic motivation by fulfilling the core psychological needs of autonomy, competence, and relatedness (Lewis, 2024; Yang et al., 2017). By providing structured challenges, immediate feedback, and opportunities for social play, gamified environments foster deep cognitive and affective engagement (Ke et al., 2016; Sailer & Sailer, 2021). Concurrently, gamification operates as a social transaction, a dynamic best understood through the lens of SET and experiential frameworks.

In this context, users implicitly weigh the perceived “costs” of learning a new system against the psychological and social “benefits” of gameplay. Framing library gamification through SET, alongside related concepts like Corporate Social Responsibility (CSR) narratives, elucidates how meaningful feedback and social-value narratives drive continued participation and value co-creation (Andre Ilyas et al., 2025; Jun et al., 2020).

Translating these theoretical insights into practice, the evidence base underscores that gamification is not a universal panacea; rather, its effectiveness relies heavily on intentional, user-centered design (Anuradhani et al., 2023). A primary design consideration is strategic onboarding, where gradual, well-structured introductions reduce initial friction and cognitive overload, smoothly transitioning users into game-based platforms or makerspace environments (Vidaković et al., 2023). Furthermore, the emotional affordances of game interactions play a critical role in sustained engagement. Utilizing effective embodied agents or immersive technologies can cultivate “flow-like” experiences that effectively capture and maintain student interest (Guo & Goh, 2016). Incorporating compelling narratives, particularly those that emphasize community contribution and social value, also acts as a critical mediator in maintaining long-term user engagement and return intentions (Jun et al., 2020).

Ultimately, while contextual variations undoubtedly influence specific outcomes, synthesizing cross-disciplinary theories with Library and Information Science (LIS) research provides a robust framework for future applications. By thoughtfully integrating intrinsic-motivation frameworks with exchange-based considerations, libraries can design gamified experiences that transcend surface-level incentives. This holistic, theory-driven approach is essential for cultivating lasting user engagement and achieving genuine, durable instructional impact in modern library environments.

Theoretical framework: SET in outreach

Successful library outreach and student engagement can be framed through SET, which suggests that social behavior is the result of an exchange process (Homans, 1958). The purpose of this exchange is to maximize benefits and minimize costs. In the context of library usage, students weigh the “cost” of time, cognitive effort, and emotional labor against the “benefit” of information retrieval or skill acquisition.

This “cost” is frequently manifested as Library Anxiety, a phenomenon where the size and complexity of the library create feelings of intimidation and inadequacy (Mellon, 1986). Traditional library instruction or orientation, often delivered via passive lectures, may be perceived by students as high-cost activities with delayed rewards, exacerbating this anxiety. Gamification alters this equation by introducing immediate, tangible rewards and lowering the psychological barrier to entry. By embedding game elements into outreach, libraries enhance the “perceived value” of the interaction, encouraging a reciprocal relationship where the library provides a dynamic experience and the student provides engagement and attention.

Furthermore, the application of SET in this study extends beyond a simple bilateral transaction of ‘time for information’ to include the social rewards inherent in collaborative engagement. As noted by Scicluna (2025), the successful initiation of a social exchange relationship depends on the development of trust and the expectation of reciprocity. In a traditional, solitary orientation, the student bears the full psychological ‘cost’ of the unknown alone. However, by utilizing a team-based gamified structure, the library facilitates a peer-to-peer exchange where students can pool their collective ‘social capital’ to navigate the environment. This shared experience lowers the individual perceived risk and shifts the exchange from a purely transactional calculation to the initial phase of a long-term social relationship, where the library demonstrates benevolence through a high-value, low-anxiety ‘ice-breaking’ activity.

Applications in libraries

Reflecting this shift toward value-added engagement, academic libraries have successfully applied gamification across orientation, instruction, and resource promotion. Research indicates a strong user preference for gamified formats over traditional methods. For example, Reed and Miller (2020) found that undergraduate students favored gamified online orientations over standard tours. Similarly, the University of Michigan's “Bibliobouts” program successfully utilized a game-based format to teach information evaluation, effectively increasing engagement and knowledge retention (Felker & Phetteplace, 2014, as cited in Haasio et al., 2021). Furthermore, experiential learning opportunities, such as escape rooms and scavenger hunts, have gained traction. These immersive environments foster problem-solving, teamwork, and information literacy skills by turning the library into a landscape of discovery (Crowe & Sclippa, 2020; Mattson, 2021).

To stimulate interaction with collections, the University of Huddersfield partnered to prototype “Librarygame,” a system that awards points for patron interaction with library materials, thereby leveraging natural competitiveness to increase circulation (Felker & Phetteplace, 2014, as cited in Haasio et al., 2021). Additionally, digital badging systems have been employed to motivate faculty and students to find research-based solutions to real-world problems (Mattson, 2021).

Beyond the academy, public libraries have integrated gamification into youth programming. Summer reading programs frequently utilize metrics, prizes, and leaderboards to foster a sense of competition (Downie & Proulx, 2022; Felker & Phetteplace, 2014, as cited in Haasio et al., 2022). Moreover, storytime sessions and the physical design of children's sections are increasingly imbued with game elements to motivate young readers (Downie & Proulx, 2022).

Gamification also extends to internal library operations. Mattson (2021) describes a rewards-based system using “library flair” (buttons) to motivate student workers during training. Similarly, Hanshew and Alley (2021) notes the use of internal games, such as adaptations of Trivial Pursuit, to refresh staff knowledge of procedures and facilitate strategic planning.

Efficacy, nuance, and challenges

The empirical evidence supporting gamification is robust, specifically regarding its ability to capture attention and influence behavior (Haasio et al., 2022). Oyinlola et al. (2024) reported that 91% of students surveyed found gamified activities fun, while 93% reported learning something new about the library. The integration of technology into gamified library services, such as mobile applications that facilitate interactive learning experiences, has also emerged as an effective strategy to reach various user groups, including first-year students (Leenaraj et al., 2023).

However, scholars emphasize that successful implementation requires more than surface-level changes. Triantafyllou et al. (2025) argue that gamification must be distinguished from “serious games” and is most effective when grounded in solid instructional design principles. They note that rather than simply transforming routine activities into games, libraries must redesign learning processes to ensure they actually meet learners' needs and expectations (Triantafyllou et al., 2025).

Despite these benefits, the implementation of gamification is not without challenges. Aminu and Shehu (2025) highlight concerns regarding competition anxiety and the potential for digital exclusion where infrastructure is lacking. Furthermore, recent literature suggests that the success of these interventions relies on the users' “game literacy”, their familiarity with game mechanics, suggesting that enhancing this literacy could further improve user experiences (Adeyemi et al., 2025). This is corroborated by Durodolu et al. (2025), who found that while perception of gamification is generally positive (60.6% approval), familiarity varies significantly by academic level. Their study of 193 information science students revealed that nearly 27% were “not familiar at all” with gamification concepts, indicating that without proper scaffolding, a gamified service might alienate a significant portion of the user base (Durodolu et al., 2025).

In summary, gamification represents an innovative approach to library services that aligns well with the principles of SET. By increasing the immediate value of library interactions through game mechanics, libraries can foster deeper engagement. However, for these strategies to remain effective, libraries must balance the benefits of motivation against the challenges of inclusivity, specifically addressing the barriers of library anxiety and game literacy.

Methodology

Research design

To investigate the immediate affective impact of gamification, this study employed a Quantitative, Quasi-Experimental, Post-Test Only Non-Equivalent Groups Design.

A post-test-only design was selected to mitigate the “testing threat” to validity; administering a pre-test on anxiety immediately before the orientation could have inadvertently primed participants to feel anxious or sensitized them to the intervention (Creswell, 2014). Instead, the design assumed that the random assignment of intact class sections to treatment groups randomized baseline anxiety sufficiently to allow for a valid post-intervention comparison.

Although a pre-test/post-test design could have established a baseline for library anxiety, a post-test-only control group design was intentionally chosen to avoid pre-test sensitization (Campbell & Stanley, 1963). The rationale was that administering an anxiety scale immediately before the orientation risked priming participants, potentially inflating baseline anxiety and altering their spontaneous engagement with the intervention.

By using the control group (which received the traditional orientation) as a benchmark for the standard post-orientation anxiety levels of first-year students, the study aimed to isolate the unique affective impact of the gamified intervention. Assuming that the intact classes began with comparable levels of general academic anxiety before the orientation, the observed significant difference in post-test scores can thus be reasonably attributed to the distinct orientation modalities used.

The study compared two distinct conditions:

- Group A (Control): Traditional Instruction (Lecture + Tour).
- Group B (Experimental): Gamified Instruction (Narrative Quest/Scavenger Hunt).

Population and sampling

The target population comprised first-year undergraduate students enrolled in mandatory introductory academic writing or university orientation courses at the University of Malta. This demographic was selected because they are the population most susceptible to "Library Anxiety" (Mellon, 1986) and are at the critical initiatory social transaction stage of the Social Exchange relationship.

- Sampling Technique: Convenience sampling was utilized by selecting existing scheduled library orientation sessions.
- Assignment: Intact classes were assigned to either Group A or Group B to minimize disruption to the academic schedule. To control for time-of-day or day-of-week effects, sessions were counterbalanced (e.g., if a Traditional session was held on Monday morning, a Gamified session was also scheduled for a Monday morning).
- Sample Size: An a priori power analysis indicated that a minimum sample size of $N = 128$ (64 per group) was required to detect a medium effect size ($d = 0.5$) with 80% power at $\alpha = 0.05$. The final sample consisted of $N = 132$ participants.

Instrumentation

Data were collected using a single, three-part printed survey administered during the final five minutes of the orientation session. The instrument operationalized the core components of Social Exchange Theory (Cost, Benefit, and User Characteristics).

Part A: perceived cost (library anxiety)

To measure the psychological "cost" of library usage, the study adapted the Library Anxiety Scale (LAS) (Bostick, 1992). To ensure the survey remained brief, a short-form, 5-item scale focusing on "Comfort with the Library" and "Staff Approachability" was used.

- Sample Item: "I feel uncomfortable asking a librarian for help."
- Scale: 5-point Likert (1 = Strongly Disagree to 5 = Strongly Agree).
- Reliability: Cronbach's alpha for the adapted scale was $\alpha = 0.82$, indicating good internal consistency.

Part B: perceived benefit (exchange value)

To operationalize the "profit" of the interaction, the survey measures the intention to return and user advocacy.

- Net Promoter Score (NPS): "On a scale of 0-10, how likely are you to recommend this library to a friend?"
  - Justification: While traditionally a corporate metric (Reichheld, 2003), NPS was selected over standard satisfaction scales because of its predictive validity regarding future behavior. Within the LIS context, Appleton (2017) argues that NPS serves as a superior proxy for user loyalty and advocacy. Unlike "satisfaction," which looks backward at a specific transaction, NPS looks forward, indicating whether the social exchange generated enough "surplus value" to compel the student to vouch for the service to peers.
- Return Intention: "I plan to visit the library voluntarily in the next month to study or find books." (Likert 1–5).

Part C: moderator (game literacy)

To address the "user heterogeneity" identified by Durodolu et al. (2025), this section assessed the participant's familiarity with gaming.

- Items: Frequency of digital gameplay (Daily/Weekly/Rarely/Never) and self-identification ("I consider myself a gamer").
- Purpose: These items were used to dichotomize the sample into "High Game Literacy" and "Low Game Literacy" subgroups for moderation analysis.

Pilot study

Prior to the main data collection, a pilot study was conducted with a convenience sample of 12 undergraduate students (who were not part of the final sample) to evaluate the playability, timing, and clarity of the gamified intervention. The primary objective was to ensure "content validity", specifically, that the game mechanics were intuitive for students with varying levels of game literacy.

Feedback from the pilot indicated that one specific puzzle involving a QR code in the stack area was "too obscure" for participants unfamiliar with scanning technology. Consequently, the game materials were refined to include explicit textual instructions alongside the QR codes. The pilot also confirmed that the 45-minute duration was sufficient for teams to complete the "quest" without rushing, ensuring that the intervention fit within the standard class allocation.

Procedure

To ensure internal validity, both the Control and Experimental sessions covered the identical learning outcomes (e.g., locating the circulation desk, finding a book by call number, accessing the printer).

1. Group A (Traditional): Students entered the library and were seated for a 15-min presentation on library rules and services. This was followed by a 25-min guided walking tour led by a librarian, who pointed out key locations.
2. Group B (Gamified): Students entered and were immediately placed into small teams. They were given a "Mission Brief" (narrative context) and a set of clues. To "escape" or "complete the mission," they were required to physically navigate to the same key locations as Group A and perform a simple task (e.g., scanning a QR code at the printer). The librarian acted as a "Game Master," providing hints rather than direct answers.
3. Data Collection: At the 30–40-min mark, both groups were gathered. The librarian distributed the survey, explaining that it was anonymous and for service improvement purposes.

Data analysis

Data was analyzed using SPSS (Version 28). The analysis proceeded in three stages aligned with the Research Questions:

- RQ1 (Anxiety): An Independent Samples *T*-Test compared the mean Library Anxiety scores of Group A vs. Group B to determine if the gamified approach successfully reduced the "cost" of entry.
- RQ2 (Exchange/NPS): A Mann-Whitney *U* Test was used to compare the Net Promoter Scores due to the non-normal distribution of the data. Pearson's Correlation tested the relationship between "Perceived Fun" and "Intention to Return."
- RQ3 (Moderation): A Two-Way ANOVA (Instructional Method X Game Literacy) was conducted to test the interaction effect, specifically examining whether non-gamers in the gamified group experienced different anxiety outcomes than gamers.

Ethical considerations

All participants were informed that the survey was voluntary and anonymous. No identifying data (names, student IDs) were collected to ensure students did not fear that their "anxiety" responses would affect their academic standing.

Funding disclosure statement

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Results

Participant characteristics

The final dataset comprised $N = 132$ first-year undergraduate students, relatively evenly distributed between the Traditional instructional condition ($n = 65$) and the Gamified condition ($n = 67$). Regarding the moderator variable, 58.3% ($n = 77$) of the sample self-identified as having "High Game Literacy" (engaging with digital games weekly or daily), while 41.7% ($n = 55$) identified as having "Low Game Literacy."

RQ1: impact on library anxiety (perceived cost)

The first research objective sought to determine if the instructional format influenced the immediate psychological state of the user. Inspecting the distribution of Library Anxiety Scale (LAS) scores revealed distinct patterns. While the Traditional Group showed a distribution centered around the mid-point of the scale ($M = 3.42$, $SD = 0.88$), the Gamified Group's distribution was skewed heavily toward the lower end ($M = 2.14$, $SD = 0.76$).

An independent samples *t*-test confirmed that this directional difference was statistically distinguishable, $t(130) = 8.42$, $p < 0.001$. However, more illustrative than the *p*-value is the standardized mean difference (Cohen's $d = 1.55$). This large effect size suggests a substantial separation between the groups, though it likely reflects the immediate nature of the post-test measurement.

Robustness Check: Given the potential non-normality often found in affective Likert data, a confirmatory Mann-Whitney *U* test was conducted. The results ($U = 654.0$, $p < 0.001$) aligned with the parametric analysis, suggesting the finding is robust across analytical assumptions.

RQ2: Social Exchange & Return Intention (perceived benefit)

Regarding the "Benefit" side of the exchange, the analysis focused on the intention to return and user advocacy.

Net Promoter Score (NPS)

The Gamified Group reported a median NPS of 9 (Promoter range), compared to a median of 6 (Detractor/Passive range) for the Traditional Group. A Mann-Whitney *U* test indicated this shift in central tendency was statistically clear ($U = 745.5$, $p < 0.001$). However, visual inspection of the frequency data shows that while the gamified group had fewer detractors, it was not devoid of passive responses, indicating that the intervention did not uniformly convert all participants into advocates.

Intention to return

The Gamified Group reported a higher mean intention to return ($M = 4.35$, $SD = 0.65$) than the Traditional Group ($M = 3.12$, $SD = 1.05$). Notably, the standard deviation in the Traditional Group (1.05) was considerably larger, suggesting a wider variance in how students respond to standard lectures, some find them helpful, while others disengage. In contrast, the lower variance in the Gamified Group suggests a more consistent, albeit specifically "novel," user experience (Table 1).

RQ3: The moderating role of game literacy

To assess whether the intervention's success was dependent on prior user characteristics, a Two-Way ANOVA was conducted.

The analysis identified a clear main effect for Instructional Method ($F(1, 128) = 65.2$, $p < 0.001$) and a secondary main effect for Game Literacy ($F(1, 128) = 4.8$, $p = 0.03$). Crucially, the interaction term was significant ($F(1, 128) = 5.92$, $p = 0.016$).

Decomposing this interaction reveals a "differential benefit":

Table 1
Comparison of affective outcomes by condition.

Outcome variable	Traditional group ($n = 65$)	Gamified group ($n = 67$)	*t*-value	*p*-value	Cohen's *d*
Library anxiety (1–5)	3.42 (0.88)	2.14 (0.76)	8.42	<0.001	1.55
Return intention (1–5)	3.12 (1.05)	4.35 (0.65)	7.81	<0.001	1.38
Net promoter score (0–10)	6.0 (Median)	9.0 (Median)	–	<0.001	–

Note: Standard Deviations appear in parentheses.

- In the Traditional Condition: Anxiety levels were statistically effectively equivalent for High Literacy ($M = 3.38$) and Low Literacy ($M = 3.46$) students. The method was equally moderately anxiety-inducing for both.
- In the Gamified Condition: While both subgroups improved, the magnitude of improvement differed. High Literacy students reported extremely low anxiety ($M = 1.85$), whereas Low Literacy students reported moderate-low anxiety ($M = 2.68$).

This indicates that while the gamified approach was universally superior to the lecture in this specific context, the "Low Game Literacy" group did not derive the same magnitude of benefit as their "High Literacy" peers, likely due to the cognitive load of learning game mechanics.

Discussion

The primary aim of this study was to examine whether gamified orientation could serve as a viable primary exchange encounter mechanism to alter the social exchange between students and the academic library. The data indicates a consistent directional trend: students exposed to the gamified condition reported lower immediate anxiety and higher return intentions than those in the traditional condition. However, the magnitude of these effects and the nuances of the moderation analysis warrant a careful, context-specific interpretation.

Contextualizing the "cost" reduction (RQ1)

The substantial effect size observed for Library Anxiety ($d = 1.55$) is notable, but it must be interpreted through the lens of immediacy and demand characteristics. In the first "trade" of time/effort for information/comfort scenarios, anxiety is often driven by the unknown (Mellon, 1986). The gamified intervention effectively replaced the "passive observer" role with an active "player" role, masking the initial intimidation of the space. Crucially, this reduction in anxiety cannot be attributed to game mechanics alone, but rather to the social architecture of the exchange. Unlike the traditional group, where students processed the library environment as solitary individuals, the gamified group operated in collaborative teams of three to five. From a Social Exchange perspective, this peer-to-peer structure allowed for a 'distribution of risk'; the psychological cost of navigating an unfamiliar 'fortress' was shared among a collective. By facilitating this shared social experience, the library lowered the perceived 'entry price,' transforming a potentially high-risk individual encounter into a low-risk social bond where students drew on each other's social capital to succeed.The observed decrease in anxiety is not solely due to the game mechanics, but fundamentally rooted in the social framework of the activity. While the control group experienced the library environment as solitary individuals, the gamified group engaged in collaborative teams of three to five. This peer-to-peer structure facilitated a "distribution of risk," viewed through the lens of SET. By sharing the psychological burden of navigating an unfamiliar “fortress” collectively, the library effectively lowered the perceived “entry price.” This transformed what could have been a high-risk individual challenge into a low-risk social bond, allowing students to leverage each other's social capital for success.

The observed significant reduction in immediate anxiety following the gamified intervention requires careful consideration regarding the ‘novelty effect’ (Koivisto & Hamari, 2014). The positive affective response and lower anxiety scores might partly stem from the temporary emotional relief and heightened engagement associated with a novel, game-based change to the typical lecture format. Therefore, longitudinal research is essential to assess whether this reduction in library anxiety is sustained after the orientation game's novelty diminishes.

The limits of “fun” in social exchange (RQ2)

The correlation between perceived fun and return intention (r = 0.72) aligns with the SET proposition that positive immediate rewards drive relationship formation (Scicluna, 2025). The shift in Net Promoter Scores suggests that for this cohort, the gamified format helped reframe the library from a purely transactional space to a relational one.

The lower satisfaction scores observed among some gamified group participants, despite their successful orientation, may stem from a mismatch between game mechanics and their expectations. Research suggests that adult learners in higher education can resist elements of ‘play,’ viewing activities like scavenger hunts as trivial and inconsistent with the ‘academic seriousness’ expected at university (Whitton, 2012). Consequently, introducing a game may violate the traditional ‘cost-benefit’ exchange these students anticipate, a straightforward transfer of information, leading to their passive engagement and lower reported satisfaction.

The “literacy tax”: An equity consideration (RQ3)

Perhaps the most ecologically valid finding of this study is the interaction effect regarding Game Literacy. While previous literature has often discussed gamification as a broad engagement tool (Felker & Phetteplace, 2014), the data here suggests a more complex reality.

The finding that “Low Game Literacy” students reported higher anxiety (M = 2.68) than “High Game Literacy” students (M = 1.85) within the same intervention points to a potential “Literacy Tax.” For non-gamers, the cognitive load was split between navigating the library and deciphering the game mechanics. This supports the cautions raised by Durodolu et al. (2025) and Triantafyllou et al. (2025). While the gamified approach was still an improvement over the traditional lecture for these students, the gap suggests that without careful design, specifically “low floor” mechanics, gamification risks introducing a new form of friction for a specific subset of the user population. Even though the game was refined during a pilot study to ensure basic usability (see Section 3.4), the results still show a gap between gamers and non-gamers. This implies that no matter how much you ‘fix’ a game, the very act of playing constitutes a cognitive load for non-gamers.

Implications for practice

These findings support the strategic implementation of gamification, subject to specific caveats:

1. Focus on Affect over Content: Gamification appears most effective for managing affect (reducing fear) during orientation. It opens the door, but it should not be assumed to replace deep information literacy instruction.
2. Inclusive Design: To mitigate the “literacy tax,” mechanics must be intuitive. The goal is to ensure that the “game” never becomes a harder barrier to scale than the library itself.

Conclusion

This study set out to interrogate the premise that gamification can effectively dismantle “Library Anxiety” by altering the perceived costs and benefits of the initial library encounter. The results offer empirical support for this hypothesis, suggesting that within the specific context of initiatory social transaction orientation, gamified interventions can significantly reduce immediate feelings of intimidation and foster a stronger willingness to return.

However, these findings should not be read as an unconditional endorsement of gamification. The study highlights that the efficacy of these interventions is bounded by user characteristics, specifically, Game Literacy. The “literacy tax” observed among non-gamers serves as a critical reminder that pedagogical innovation must remain sensitive to equity. If the mechanics of the intervention are too complex, the library risks replacing one form of anxiety with another.

Limitations and future directions

Several methodological limitations must be acknowledged when interpreting the current findings. A primary constraint is the absence of an initial baseline measure for library anxiety, due to the study's necessary use of a post-test-only design to prevent pre-test sensitization. Consequently, while the data establishes comparative differences between the two orientation methods, the absolute reduction in anxiety from an individual student's starting point cannot be quantified.

A further limitation relates to the timing of measurement: outcomes were assessed immediately following the intervention in a controlled environment. This makes it challenging to determine whether the lower anxiety scores observed in the gamified group represent a lasting cognitive shift or are merely a reflection of the momentary positive affect and emotional relief generated by the game. Furthermore, the strong positive results may have been influenced by the novelty of the experience and demand characteristics related to the researchers' presence.

To address these methodological shortcomings, future research should adopt Solomon four-group designs. This approach would allow for safely establishing baseline measures without priming participants. Additionally, longitudinal studies are necessary to evaluate whether the initial anxiety reduction persists over a period of weeks or months, and, crucially, if it ultimately leads to sustained, independent information-seeking behavior throughout the academic year.

Final thought

Ultimately, this research suggests that the value of gamification in academic libraries lies not in the mechanics of points or badges, but in its capacity to humanize the institution. By temporarily suspending the rules of the “fortress” and inviting students to play, libraries can create a lower-stakes environment for that critical first interaction. At the end of the day, games aren't magic, but if they help a scared first-year student feel like they belong in the library, they are worth the effort.

CRediT authorship contribution statement

Ryan Scicluna: Writing – original draft.

Ethical approval and informed consent

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee. Ethical approval was obtained from the relevant research ethics committee at the University of Malta. Informed consent was obtained from all individual participants included in the study; all participants were explicitly informed that the survey was voluntary and anonymous. To protect participant anonymity and ensure no perceived risk to academic standing, no identifying data (such as names or student IDs) were collected.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Google Gemini to assist in the instructional design of the gamified intervention and to refine the structure and clarity of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

The author(s) declare(s) that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Post-Orientation Survey Instrument

Instructions: Thank you for participating in today's library orientation. Please answer the following questions honestly. Your responses are **anonymous** and will not affect your grades.

Part 1: About You

Please tick the box that best describes you.

1 How often do you play digital games (Mobile, Console, or PC)?
- [ ] Daily
- [ ] Weekly
- [ ] Monthly
- [ ] Rarely / Never

2 Do you consider yourself a "Gamer"?
- [] Yes
- [] No

Part 2: Your Experience Today

Please circle the number that best matches your feeling right now.

Statement	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1. I feel comfortable being in the library.	1	2	3	4	5
2. I would feel nervous asking a librarian for help.	1	2	3	4	5
3. The library feels like an intimidating place to me.	1	2	3	4	5
4. I feel confident I can find what I need here.	1	2	3	4	5
5. I feel like I don't know what to do in the library.	1	2	3	4	5

(1 = Strongly Disagree, 5 = Strongly Agree)

Part 3: Feedback

1 I plan to visit the library voluntarily in the next month to study or borrow books.
- [] Strongly Disagree
- [] Disagree
- [] Neutral
- [] Agree
- [] Strongly Agree

2 On a scale of 0 to 10, how likely are you to recommend this library orientation to a friend?
- [0] [1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
- *(0 = Not at all likely, 10 = Extremely likely)*

Appendix B. Instructional Protocols

1. The experimental intervention (gamified group)

Activity Title: *Operation: Codex Recovery* **Format:** Narrative Scavenger Hunt / Escape Room Lite **Duration:** 45 Minutes **Team Size:** 3–5 Students per team.

The Mission Brief (Read aloud to students):

"Welcome, agents. You are standing in the central archive of human knowledge. However, a digital virus has scrambled our catalog. We have lost the location of the *'Master Index.'* Your team has 30 minutes to recover four encryption keys hidden throughout the building. If you fail, the knowledge stored here will be lost forever. Your clues are in this envelope. Good luck."

The Game Flow (The 4 Quests):

- Quest 1: The Sentinel (Circulation Desk)

- *Clue:* "I stand guard at the entrance, checking every book that leaves. Speak to me, the Sentinel (Librarian at the Desk) and let me show you how to find your way inside the Library"
  - *Task:* Students must approach the circulation desk and ask a reference question. This could be how to find resources about a specific topic so students need to be specific with their question so that the Librarian can show them how to use HyDi - the Library search portal.
  - *Reward:* The Librarian hands them a puzzle piece (Encryption Key 1).
  - *Learning Outcome:* Reduces fear of speaking to staff and learn about HyDi.
- Quest 2: The Labyrinth (The Stacks)
  - *Clue:* "Find the book written by [Author Name] published in [Year]. Inside, you will find the next code."
  - *Task:* Students must use the catalog (HyDi) to find the call number (Shelfmark), locate the shelf, and open the specific book.
  - *Reward:* A bookmark hidden inside the book with a QR code (Encryption Key 2).
  - *Learning Outcome:* Catalog searching and Call Number navigation.
- Quest 3: The Replicator (Printers/Copiers)
  - *Clue:* "To save the data, you must know how to duplicate it. Scan this QR code at the Student Printing Station."
  - *Task:* Students locate the printer room and scan a poster.
  - *Reward:* A digital code pops up on their phone (Encryption Key 3).
  - *Learning Outcome:* Location and use of IT facilities.
- Quest 4: The Digital Uplink (Online Resources)
  - *Clue:* "The physical archives are compromised. You must access the secure cloud server. On your mobile device, go to the Library Homepage. Click on 'Databases A-Z' and find 'JSTOR'. The Encryption Key is the **first word** of the database description."
  - *Task:* Students must successfully navigate the mobile library website, locate the specific database link, and read the description to find the password.
  - *Reward:* They report the password to the Game Master to receive the Final Key.
  - *Learning Outcome:* Navigating the library website and accessing e-resources.

Victory Condition: Teams return to the "Game Master" with all 4 keys. They receive a small prize (e.g., a sticker or candy) and the "Game Debrief" survey.

2. The control intervention (traditional group)

Activity Title: *Library Orientation Seminar & Tour* **Format:** Lecture + Guided Walk **Duration:** 45 Minutes.
Protocol:
Phase 1: The Presentation (15 Minutes)

- Students are seated in the library seminar room.
- Librarian uses a PowerPoint presentation covering:
  - Library opening hours.
  - Rules of conduct (Silence in study zones, no food).
  - Different study spaces and departments.
  - How to use the Library resources (HyDi).
  - Borrowing policies and fines.

Phase 2: The Guided Tour (25 Minutes)

- Librarian leads the group through the library.
- **Stop 1 (Circulation Desk):** (Matches Quest 1).
- **Stop 2 (The Shelves - Main Collection):** (Matches Quest 2).
- **Stop 3 (The study spaces - IT Facilities):** (Matches Quest 3).
- **Stop 4 (Random space in the Library - Online resources):** (Matches Quest 4).

Phase 3: Conclusion (5 Minutes)

- Group returns to the seminar room.
- Librarian asks if there are any questions (Q&A).
- Survey is distributed.

Data availability

Data will be made available on request.

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